

Time-series momentum in the Chinese commodity futures market

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Abstract

This study examines time-series momentum in the Chinese commodity futures market. The findings show that a time-series momentum strategy performs best with a one-month look-back period and a one-month holding period. Furthermore, this strategy outperforms passive long and cross-sectional momentum strategies in the Chinese futures market based on Sharpe ratios, risk-adjusted excess returns, and cumulative returns. But highly volatile market characteristic with many speculative investors limits the period in which time-series momentum is maintained. Our findings suggest that the anomaly is observed in international asset markets, including Chinese commodity futures, and support the implication that speculators profit from time-series momentum strategy is the expense of hedgers.

Keywords

China; Commodity; Futures; Momentum; Time-series

JEL Classification

G12, G13, G15, F37

1 | INTRODUCTION

Momentum strategies, which are representative anomalies of financial markets, have been actively researched in relation to equity markets. Momentum strategies are divided into two broad groups, namely, cross-sectional momentum and time-series momentum strategies. In the last decade,

numerous studies around the world have examined momentum in futures markets in order to show the significance of momentum strategies, predominantly focusing on cross-sectional momentum. But, recently, several studies on time-series momentum in futures markets suggest that a time-series momentum strategy outperforms a cross-sectional momentum strategy in developed countries.

Many of studies have been extended to examine momentum in futures markets around the world in order to show the significance of momentum strategies but mainly focused on cross-sectional momentum strategy. A cross-sectional momentum strategy assumes that those equities with the highest average returns in the recent past, the so-called winners, outperform equities with the worst returns, the so-called losers. Jegadeesh and Titman (1993) show that a cross-sectional momentum strategy yields significant excess returns in the equity markets. Recently, numerous studies apply momentum strategies to commodity futures markets. Miffre and Rallis (2007), Shen, Szakmary, and Sharma (2007), Fuertes, Miffre, and Rallis (2010), Asness, Moskowitz, and Pedersen (2013), Blitz and De Groot (2014), Clare, Seaton, Smith, and Thomas (2014), Szymanowska, Roon, Nijman, and Van den Goorbergh (2014), and Bakshi, Gao, and Rossi (2018) study cross-sectional momentum in commodity futures for different periods, although with different definitions of winners and losers. According to Miffre (2016), the average Sharpe ratio of a cross-sectional momentum strategy is 0.5, which is much greater than the value of -0.24 for a long-only equally weighted portfolio. In addition, according to Kang and Kwon (2017), a cross-sectional momentum strategy cannot be fully explained by traditional risk factors, macroeconomic variables, and commodity-sector momentum in the futures market. This suggests that momentum also prevails in commodity futures markets.

In contrast, Time-series momentum in futures markets is a relatively recent topic of research and received little academic attention compared with studies on cross-sectional momentum in spite of its economic implications. Unlike a cross-sectional momentum strategy, a time-series momentum strategy focuses on the past returns of single commodity futures. The strategy suggests taking a long position if the commodity has had positive returns in the past, and a short position if it has had negative returns. Furthermore, time-series momentum considers absolute returns, not relative returns as cross-sectional momentum does, and is an anomaly related to the autocorrelation and continuation of returns that is counter to the random-walk hypothesis. Several studies on time-series momentum in futures markets have found that time-series momentum strategies outperform cross-sectional momentum strategies. Moskowitz, Ooi, and Pedersen (2012) document the significance of time-series momentum in equity index, currency, commodity, and bond futures in developed countries such as the United States, the United Kingdom, Australia, New Zealand, Japan, France, Germany, Italy, Netherlands, Spain, and other European countries. Their research is based on 58 instruments for the period January 1965 to December 2009. They use the most liquid contracts, which are typically the nearest or the next-nearest to the delivery contract. They show that a time-series momentum strategy outperforms both a passive long strategy and a cross-sectional momentum strategy using a regression

analysis with benchmarks such as the MSCI World Equity Index, Barclay's Aggregate Bond Index, S&P GSCI Index, long-short factors, and time-series momentum returns. Small minus big (SMB), high minus low (HML), and up minus down (UMD) are considered as the long-short factors. According to Kenneth French's data library,¹ the SMB is "the average return on the three small portfolios minus the average return on the three big portfolios," the HML is "the average return on the two value portfolios minus the average return on the two growth portfolios," and the UMD is "the average return on the two high prior return portfolios minus the average return on the two low prior return portfolios."

Moskowitz, Ooi, and Pedersen (2012) also determine the relationship between the time-series momentum and cross-sectional momentum, as well as the predictability of positions, price changes, and the roll yield. Furthermore, Hurst, Ooi, and Pedersen (2013) show that a time-series momentum strategy can explain the returns of managed futures funds. Hurst, Ooi, and Pedersen (2017) extend the time-series momentum strategy to the global market and longer sample periods. Kojen, Moskowitz, Pedersen, and Vrugt (2018) analyze carry trades with time-series momentum returns as a risk factor. Gao, Han, Li, and Zhou (2018) introduce intraday momentum in the market using exchange-traded fund (ETF) data. However, Goyal and Jegadeesh (2018) argue that a cross-sectional strategy significantly outperforms a time-series strategy after risk adjustment. But the existing literatures of time-series momentum in commodity futures have been researched mainly focusing on the U.S. markets.

In this study, we investigate the time-series momentum with an international perspective, focusing on the Chinese commodity futures market and examine the significance of time-series momentum. China's recent rapid economic development² has made it a major participant in world commodity markets (Roache, 2012). Figure 1 shows the number of traded derivatives contracts by country as of 2016,³ indicating that China's trading volume of derivatives is the second largest in the world. The Chinese market will continue to grow, in which case, the international commodity market will become more dependent on the Chinese market, indicating a shift in the market trend dominated by the United States market in the past. Therefore, we investigate the Chinese commodity futures market and, specifically, the time-series momentum, an anomaly observed in various asset markets.

[Figure 1]

¹ http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/f-f_factors.html.

² Su and Liu (2016) note that human capital and foreign direct investment are major contributors to Chinese economic growth.

³ The data are collected from Statista, "Largest derivatives exchanges worldwide in 2016, by number of contracts traded (in millions)" (<https://www.statista.com/statistics/272832/largest-international-futures-exchanges-by-number-of-contracts-traded/>).

We examine time-series predictability for Chinese commodity futures and the significance of the time-series momentum strategy from the regressions of its returns on the risk factors, with different look-back and holding periods. We also compare the performance of time-series momentum strategy with those of cross-sectional and passive long strategies. Our results suggest that the most recent 12 months have shown the strongest and most significant return continuation. For comparison with U.S. futures market, the time-series momentum strategy is constructed following the methodology of Jegadeesh and Titman (1993). In addition, our results reveal that a strategy employing a one-month look-back period and holding period performs best, similarly to the results for a cross-sectional momentum strategy obtained by Shen, Szakmary, and Sharma (2007) and Kang and Kwon (2017) in their research on U.S. and Chinese commodity futures, respectively. In order to examine the robustness of our findings, we evaluate the time-series momentum strategy using the Sharpe ratio, risk-adjusted excess returns, and cumulative returns. Then, we compare the results to those of a passive long strategy and a cross-sectional strategy based on the methodology of Asness, Moskowitz, and Pedersen (2013). Overall, the time-series momentum strategy outperforms the latter two strategies, in general, although not for agriculture commodities, which also have relatively lower Sharpe ratios in the U.S. commodity futures market. This study contributes existing literature on the futures market, since the Chinese futures market has become more influential in international futures markets.

This paper is organized as follows. Section 2 describes our data, and Section 3 explains the methodologies used for our analysis. Section 4 presents the empirical results on the time-series momentum strategy. Lastly, Section 5 concludes the paper.

2 | DATA

The data period begins from September 14, 2006, when at least 10 of the nearest-to-delivery commodity futures contracts (the most liquid contracts) started being traded actively in the Chinese market. The daily closing prices of the generic first futures (in USD) between September 14, 2006, and October 31, 2017, are collected from Bloomberg (see Table 1).

[Table 1]

Bloomberg provides 44 generic first Chinese commodity futures. Generic first futures refers to futures that are the nearest to the delivery contract. Trading in aluminum generic first futures started in 2004, copper cathode in 1999, corn in 2004, fuel oil in 2004, natural rubber in 2002, Zhengzhou cotton in 2006, no. 1 soybeans in 2002, soybean meal in 2000, and white sugar in 2006. Because Zhengzhou cotton has been traded actively since September 14, 2006, the data period is set from September 14, 2006, to October 31, 2017.

Table 2 provides the summary statistics (annualized arithmetic means and annualized volatilities) of the daily excess returns on the futures contracts referred to in Table 1. The average annualized mean and volatility of the Chinese commodity futures are 5.37% and 24.43%, respectively, which are similar, on average, to those of the U.S. commodity futures, which are 5.47% and 23.43%, respectively (Moskowitz, Ooi, & Pedersen, 2012).

[Table 2]

Table 3 shows the correlations between the Chinese commodity futures for the sample period January 2007 to October 2017. The average correlation between these futures is 0.24. The correlations between copper cathode and aluminum and between natural rubber and copper cathode are the highest, at 0.56, and the correlation between fuel oil and corn is the lowest, at -0.16. The low correlations between commodity futures is controversial, with Erb and Harvey (2006) indicating that commodity futures have low correlations in the U.S. market. Similarly, low correlations are evident in the Chinese commodity futures market, as shown in Table 3.

[Table 3]

Moskowitz, Ooi, and Pedersen (2012) utilize the MSCI World Equity Index and long-short factors to evaluate the returns of the strategies. The MSCI World Equity Index represents large and mid-cap equity in 23 developed markets (see Figure 2 for the sector and country weights of the index). Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hong Kong, Ireland, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, the United Kingdom, and the United States are considered developed countries.⁴ In addition, the long-short factors (the three Fama–French global factors) are obtained from Kenneth French’s data library. In addition to the United States, the regions covered by these factors include Hong Kong, Japan, and other developed countries. Because the MSCI World Equity Index and long-short factors exclude the Chinese equity markets, we also consider the CSI 300 Index, which represents capitalization-weighted Chinese stocks.

[Figure 2]

3 | METHODOLOGY

⁴ <https://www.msci.com/documents/10199/178e6643-6ae6-47b9-82be-e1fc565ededb>.

Following Moskowitz, Ooi, and Pedersen (2012), we calculate the monthly returns for each commodity as the cumulative returns from the first to the last day of the month. Figure 3 shows a graph of the monthly cumulative returns for the Chinese commodity futures. Because the volatilities are different among the commodity futures, the returns are scaled by their volatility. The ex-ante annualized variance σ_t^s for each commodity s is calculated using exponential weights, as follows:

$$\sigma_t^s = 261 \sum_{i=0}^{\infty} (1 - \lambda) \lambda^i (r_{t-1-i}^s - \bar{r}_t^s)^2, \quad (1)$$

where 261 is a scale factor used to annualize the variance, the exponential smoothing factor λ meets the condition $\sum_{i=0}^{\infty} (1 - \lambda) \lambda^i = 1$, the “center of mass” which calculated as $\sum_{i=0}^{\infty} (1 - \lambda) \lambda^i i = \lambda / (1 - \lambda)$ is 60 days, and $\bar{r}_t$ is an exponentially weighted average return with same smoothing factor λ .

[Figure 3]

The excess return r_t^s for commodity s in month t is regressed on its excess return, lagged by h months, in order to examine the predictability of the futures returns across time horizons. Both excess returns are placed on the same scale by dividing them by their ex-ante volatilities, σ_{t-1}^s , computed using equation (1). Panel A of Figure 4 shows the t -statistics from the following pooled panel regression:

$$\frac{r_t^s}{\sigma_{t-1}^s} = \alpha + \frac{\beta_h r_{t-h}^s}{\sigma_{t-h-1}^s} + \varepsilon_t^s. \quad (2)$$

In addition, the excess return r_t^s is regressed on the sign of the past excess return, lagged by h months, in order to examine the time-series momentum in a different way:

$$\frac{r_t^s}{\sigma_{t-1}^s} = \alpha + \beta_h \text{sign}(r_{t-h}^s) + \varepsilon_t^s. \quad (3)$$

In the specification in Equation (3), $\text{sign}(r_{t-h}^s)$ is +1 or −1 if r_{t-h}^s is positive or negative, respectively. Panel B of Figure 4 plots the t -statistics from the pooled panel regression.

[Figure 4]

Both regressions use lags from 1 to 60 months, and the sample period is September 2006 to October 2017. The time-series predictability for the Chinese commodity futures is indicated in Figure 4. The monthly excess returns of each futures contract, scaled by their ex-ante volatilities, are regressed on their lagged scaled excess returns and the sign of the past excess return. Panel A indicates that the regression results use the excess returns themselves as dependent variables, and that the dependent variables presented in Panel B indicate the signs of the lagged excess returns. A positive t -statistic indicates a return continuation. Figure 4 shows that both Panels A and B have significant positive t -statistics for the first year (first 12 months), and negative t -statistics for longer horizons. This result suggests that the most recent 12 months indicate the strongest and most significant return continuation, supporting the results for futures markets obtained by Moskowitz, Ooi, and Pedersen (2012). Moreover, the first month has a highly significant and positive t -statistic, suggesting a strong return continuation.

4 | EMPIRICAL RESULTS

4.1 | Time-Series Momentum

The monthly returns of the time-series momentum strategy for each commodity s are calculated following the methodology of Jegadeesh and Titman (1993). For each commodity s and month t , if the cumulative return for the past k months, from month $t-k$ to month $t-1$ (the look-back period), is positive or negative, then the value of $1/\sigma_{t-1}^s$ of the long or short position is taken and held for h months (the holding period). The number of portfolios held at month t is h , except for the first $h-1$ months. The return of commodity s at time t in the time-series momentum strategy is the average return of all portfolios at time t , where the return at time t , $r_t^{TSMOM(k,h)}$, is defined as the average return across all commodities. The look-back periods and holding periods in the strategy vary from 1, 3, 6, 9, 12, 24, 36, and 48 months. To evaluate the performance of the strategy, the following regression is conducted for each k and h :

$$r_t^{TSMOM(k,h)} = \alpha + \beta_1 MKT_t + \beta_2 BOND_t + \beta_3 GSCI_t + sSMB_t + hHML_t + mUMD_t + \varepsilon_t, \quad (4)$$

where MKT is the excess return on the MSCI World Index, $BOND$ is the excess return on the Barclay's Aggregate Bond Index, $GSCI$ is the excess return on the S&P GSCI Index, and the long-short factors (SMB , HML , and UMD) are the Fama–French (1993) factors for the size, value, and cross-sectional momentum premiums of Carhart (1997).

China became a major participant in world commodity markets because of its recent rapid economic development. As China was developed as a country which has the second largest trading volume of derivatives in the world, the Chinese futures market also became large enough to influence the world market one another. Therefore, we looked the broader market that can represent the Chinese

futures market, which is the world index. Additionally we considered CSI300 as market index for and for localization.

Panel A of Table 4 indicates the t -statistics of the alphas from the regressions stated in equation (4), for different look-back and holding periods. Compared with the U.S. commodity futures market, the Chinese commodity futures market shows less significance in the excess returns of the time-series momentum strategy. The strategies with a one-month look-back and a one-month holding period and those with a 12-month look-back and a one-month holding period have significant and positive t -statistics for risk-adjusted excess returns (2.60 and 2.42, respectively). As shown, the strategy with a one-month look-back period and holding period has the highest t -statistic in the Chinese commodity futures market. However, in the U.S. market, the strategy with a look-back period of 12 months and a holding period of one month has the highest t -statistic.

[Table 4]

In Panel B of Table 4, the same regression is conducted using the CSI 300 Index instead of the MSCI World Equity Index. As noted earlier, we use the CSI 300 Index because the MSCI World Equity Index only represents large and mid-cap equity in 23 developed markets, and excludes China. As in the regression with the MSCI World Equity Index, the strategies with a one-month look-back and holding period and those with a 12-month look-back period and a one-month holding period have significant and positive t -statistics for risk-adjusted excess returns (2.56 and 1.98, respectively). The excess returns are smaller than those in the regression based on the MSCI World Equity Index, indicating that the CSI 300 better explains the returns from the time-series momentum strategy. However, because both indices represent equity markets, not commodity markets, switching from the MSCI World Equity Index to the CSI 300 Index is not considered a significant transition. Overall, the longer holding period performs poorly for both the U.S. and Chinese commodity futures markets, suggesting that a one-month holding period performs best, in general.

Table 4 shows that the time-series momentum strategy with a one-month look-back period performs best. This is shown by the high time-series predictability with a one-month lag, which indicates a significant return continuation in one month. Because the Chinese commodity futures market is highly volatile, with many speculative investors, the time-series momentum is maintained for a short period. As Moskowitz, Ooi, and Pedersen (2012) point out, speculators who earn from a time-series momentum strategy do so at the expense of hedgers. The more speculators that participate, the less profit a time-series momentum strategy generates. The results for the strategy with a one-month look-back and holding period corresponds to those of Shen, Szakmary, and Sharma (2007), who show that a one-month cross-sectional momentum strategy for 28 commodities in the U.S. market earned the most significant profits, on average, from 1959 to 2003. Furthermore, the results

mirror those of Kang and Kwon (2017), who also examine a cross-sectional momentum strategy. They show that the strategy with a one-month look-back period and a one-month holding period has the highest return and Sharpe ratio in the Chinese commodity futures market. This result suggests that strategies with one-month look-back and holding periods perform best in the Chinese futures market, which is relatively shorter than those in the United States. Therefore, the return continuation in the Chinese futures market lasts about one month, but is longer in the U.S. market.

The time-series momentum factor is constructed following Moskowitz, Ooi, and Pedersen (2012) and the conventions used in other studies on cross-sectional momentum. The factor, called TSMOM, uses a 12-month look-back period and one-month holding period (i.e., $k=12$ and $h=1$). For comparison with U.S. futures market, we followed the convention. The TSMOM return for each commodity futures contract s at time t is given as follows:

$$r_{t,t+1}^{\text{TSMOM},s} = \text{sign}(r_{t-12,t}^s) \frac{40\%}{\sigma_{t-1}^s} r_{t,t+1}^s, \quad (5)$$

where $r_{t,t+1}^{\text{TSMOM},s}$ is the return in month t . To ensure consistency with the previous process in this study, σ_{t-1}^s is used instead of σ_t^s , following Moskowitz, Ooi, and Pedersen (2012). The overall return $r_{t,t+1}^{\text{TSMOM}}$ across all the commodity futures S_t is defined as follows:

$$r_{t,t+1}^{\text{TSMOM}} = \frac{1}{n(S_t)} \sum_{s=1}^{S_t} \text{sign}(r_{t-12,t}^s) \frac{40\%}{\sigma_{t-1}^s} r_{t,t+1}^s, \quad (6)$$

where $n(S_t)$ is the number of all commodity futures contracts, and 40% is used to make it easier to accurately compare the portfolios with those in the literature.

4.2 | Time-Series and Cross-Sectional Momentum

The cross-sectional momentum strategy considers all commodity futures at a point in time and compares the past returns across the commodity futures. In contrast, the time-series momentum strategy focuses on the past returns of one of the commodity futures. The cross-sectional momentum strategy in this study is constructed following the methodology of Asness, Moskowitz, and Pedersen (2013). They compare and rank the past 12-month returns of the instruments. The position for each commodity future at a point in time is held in proportion to its rank relative to the median rank (Moskowitz, Ooi, & Pedersen, 2012).

The cross-sectional momentum return for each commodity futures contract s at time t is given as follows:

$$r_{t,t+1}^{\text{XSMOM},s} = \frac{\text{rank}(r_{t-12,t}^s)}{2.5} \frac{40\%}{\sigma_{t-1}^s} r_{t,t+1}^s, \quad (7)$$

where $rank(r_{t-12,t}^s)$ is the rank of the past returns for the commodity futures minus the median rank. For comparison purposes, the position of the cross-sectional momentum strategy returns for a commodity $r_{t,t+1}^{XSMOM,s}$ is set to $\frac{40\%}{\sigma_{t-1}^s}$, which is the same as that for the time-series momentum for each commodity future. The scale factor of 2.5 is used to make $rank(r_{t-12,t}^s)/2.5$ 1 or -1, on average, because the time-series momentum factor has a coefficient of 1 or -1 for $(40\%/\sigma_{t-1}^s)r_{t,t+1}^s$. Here, $rank(r_{t-12,t}^s)$ is defined as the rank minus the median rank. Assume that aluminum and copper cathode have scaled excess returns of -0.59 and 1.39, and ranks of 5 and 1, respectively, on January 1, 2008. The value of $rank(r_{t-12,t}^s)$ for aluminum and copper cathode should be -0.5 and -4.5, respectively, and the median rank is 5.5 when the number of futures is 10. As with the time-series momentum strategy, the overall returns for the cross-sectional momentum strategy are calculated using equation (8).

$$r_{t,t+1}^{XSMOM} = \frac{1}{n(S_t)} \sum_{s=1}^{S_t} \frac{rank(r_{t-12,t}^s)}{2.5} \frac{40\%}{\sigma_{t-1}^s} r_{t,t+1}^s, \quad (8)$$

where $n(S_t)$ is the number of all commodity futures contracts.

A regression analysis is conducted to determine the relationship between the time-series momentum strategy and cross-sectional momentum strategy. The returns from the time-series momentum strategy are regressed on the returns from the cross-sectional momentum strategy for the sample period of January 2008 through October 2017. The results are reported in Table 5. As Table 5 shows, the t -statistic of the coefficient of the returns from the time-series momentum strategy is 5.24, which suggests that the returns are significantly related to those from the cross-sectional momentum strategy. The coefficient of determination is 0.20.

[Table 5]

These results suggest that the time-series momentum strategy outperforms the cross-sectional momentum strategy. The intercept from the regression of the returns from the time-series momentum strategy on the returns for the cross-sectional momentum strategy has a t -statistic of 2.29, which is significant.

4.3 | Performance Evaluation of the Time-Series Momentum Strategy

Figure 5 shows the annualized Sharpe ratios by commodity, and reports the annualized Sharpe ratios of the time-series momentum strategy (panel A), passive long strategy (panel B), and cross-sectional momentum strategy (panel C) by commodity futures contract for the sample period January

2008 to October 2017. Unlike the time-series momentum strategy in the U.S. commodity futures market, where all Sharpe ratios are positive, six of the 10 Chinese commodity futures have positive values and four are negative.

[Figure 5]

The Sharpe ratios of the time-series momentum strategy are greater than those of the passive long strategy and cross-sectional momentum strategy (Panels B and C of Figure 5, respectively), in general. However, the Sharpe ratio of the time-series momentum strategy is worse than that of the passive long strategy for agricultural commodity futures. A comparison of Panels A to C of Figure 5 also suggests that the time-series momentum strategy has greater Sharpe ratios than those of the cross-sectional momentum strategy in all Chinese commodity futures contracts, except for futures based on soybean byproducts (soybean meal and soybean oil). This poor performance of the time-series momentum strategy in agricultural commodity futures is also shown in the study of Moskowitz, Ooi, and Pedersen (2012). These results suggest that the time-series momentum strategy performs worse in the Chinese commodity futures market compared to in the U.S. market. However, it performs better in terms of the Sharpe ratio than the passive long strategy and the cross-sectional momentum strategy, except for agricultural commodities, as in the U.S. commodity futures market.

As risk factors, we use the GSCI index excess returns, SMB, HML, and UMD, on which we regress the returns of the time-series momentum strategy.⁵ The results of the regression analysis for the monthly returns of each strategy—the time-series momentum strategy, passive long strategy, and cross-sectional strategy—on the traditional risk factors are indicated in Panels A, B, and C of Table 6, respectively. The sample period is January 2008 to October 2017.

[Table 6]

As Table 6 shows, only the time-series momentum strategy has significant excess returns after considering the risk factors. The intercept of the regression for the time-series momentum strategy has a *t*-statistic of 2.33, while those for the passive long strategy and cross-sectional strategy have *t*-statistics of 0.04 and 0.24, respectively. These results indicate that the time-series momentum strategy performs better than the passive long strategy and cross-sectional strategy in terms of the risk-adjusted excess returns. Furthermore, we perform a regression analysis for the commodity futures, other than agricultural commodities, that have a negative Sharpe ratio. As in the previous regression analysis, the sample period is January 2008 to October 2017, and the GSCI index, SMB, HML, and UMD are used

⁵ The details are explained in Section 1.

as risk factors. Compared with the results in Panel A-1 of Table 6, those in Panel A-2 of Table 6 show that the t -statistic of the intercept increases from 2.33 to 3.11.

Another way to evaluate the performance is to examine the cumulative returns. Figure 6 shows the cumulative returns of the time-series momentum strategy, passive long strategy, and cross-sectional momentum strategy from January 2008 to October 2017. The figure shows that the cumulative return of the time-series momentum strategy is much greater than those of the passive long strategy and cross-sectional momentum strategy, suggesting that the time-series momentum strategy outperforms the other two strategies.

[Figure 6]

Table 7 shows the correlations between the returns of each Chinese futures contract in the time-series momentum strategy and passive long strategy. The overall correlation between the returns of each commodity is 0.03 for the time-series momentum strategy and 0.22 for the passive long strategy. This result suggests that the time-series momentum strategy can reduce the correlations between the Chinese commodity futures better than the passive long strategy can.

[Table 7]

5 | CONCLUSION

Our study examines the time-series momentum in the Chinese commodity futures market. The results show that the strategy with one-month look-back and holding periods perform best, which is similar to the results for the cross-sectional momentum strategy obtained by Shen, Szakmary, and Sharma (2007) and Kang and Kwon (2017) in their research on U.S commodity futures and Chinese commodity futures, respectively. These results support the initial under-reaction and delayed over-reaction theories, as noted by Moskowitz, Ooi, and Pedersen (2012). However, the momentum return is maintained for a shorter period than in the U.S. market, because the latter has far more speculative investors. This result also supports the implication that speculators profit from time-series momentum strategy is the expense of hedgers.

For comparison with U.S. futures market, we defined time-series momentum strategy with 1-month look-back and 1-month holding periods although the strategy with one-month look-back and holding periods perform best. In the Chinese commodity futures market, the time-series momentum strategy outperforms the passive long and the cross-sectional momentum strategies constructed following the process suggested by Asness, Moskowitz, and Pedersen (2013). The regression of the returns from the time-series momentum strategy on the returns from the cross-sectional momentum strategy verifies that the former outperforms the latter. Furthermore, the Sharpe ratios, risk-adjusted

performance, cumulative returns, and correlation structures of the time-series momentum strategy are compared with those of the passive long and cross-sectional momentum strategies. In general, the Sharpe ratios of Chinese commodity futures are greater than those of the other strategies. The time-series momentum strategy for all commodity futures except the agricultural commodity futures (e.g., corn, no. 1 soybeans, soybean meal, and soybean oil) performs better than the passive long strategy and cross-sectional momentum strategy. In terms of the risk-adjusted performance, the GSCI index, SMB, HML, and UMD are used as risk factors, and only the excess return of the time-series momentum strategy has a significant positive value among the three strategies. The cumulative return of the time-series momentum strategy is also much greater than those of the other two strategies. In addition, by comparing the correlation structures, we confirmed that the time-series momentum strategy can reduce the correlations between the Chinese commodity futures better than the passive long strategy can.

This study is limited by the scarcity of data and research for Chinese commodity futures. The latter market commenced trading relatively recently; hence, the period and the number of commodity futures are still insufficient. There is also a lack of research on momentum in futures markets. In particular, studies on time-series momentum in futures markets are relatively recent and, thus, few have been conducted.

Momentum is an anomaly that prevails in financial markets around the world. Further studies that analyze momentum are needed, as are studies that apply the various momentum strategies to commodity futures markets and develop better strategies. Moreover, further research is needed to investigate the reasons behind the better performance of the time-series momentum strategy compared with that of the cross-sectional momentum strategy, as well as to identify the kinds of commodity futures that perform better or worse, especially for the time-series momentum strategy.

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FIGURE 1 Number of traded derivatives contracts by county.

The number of traded derivative contracts by country as of 2016 is reported. The data on the trading volumes of the top 20 exchanges around the world are collected from Statista and classified by country.

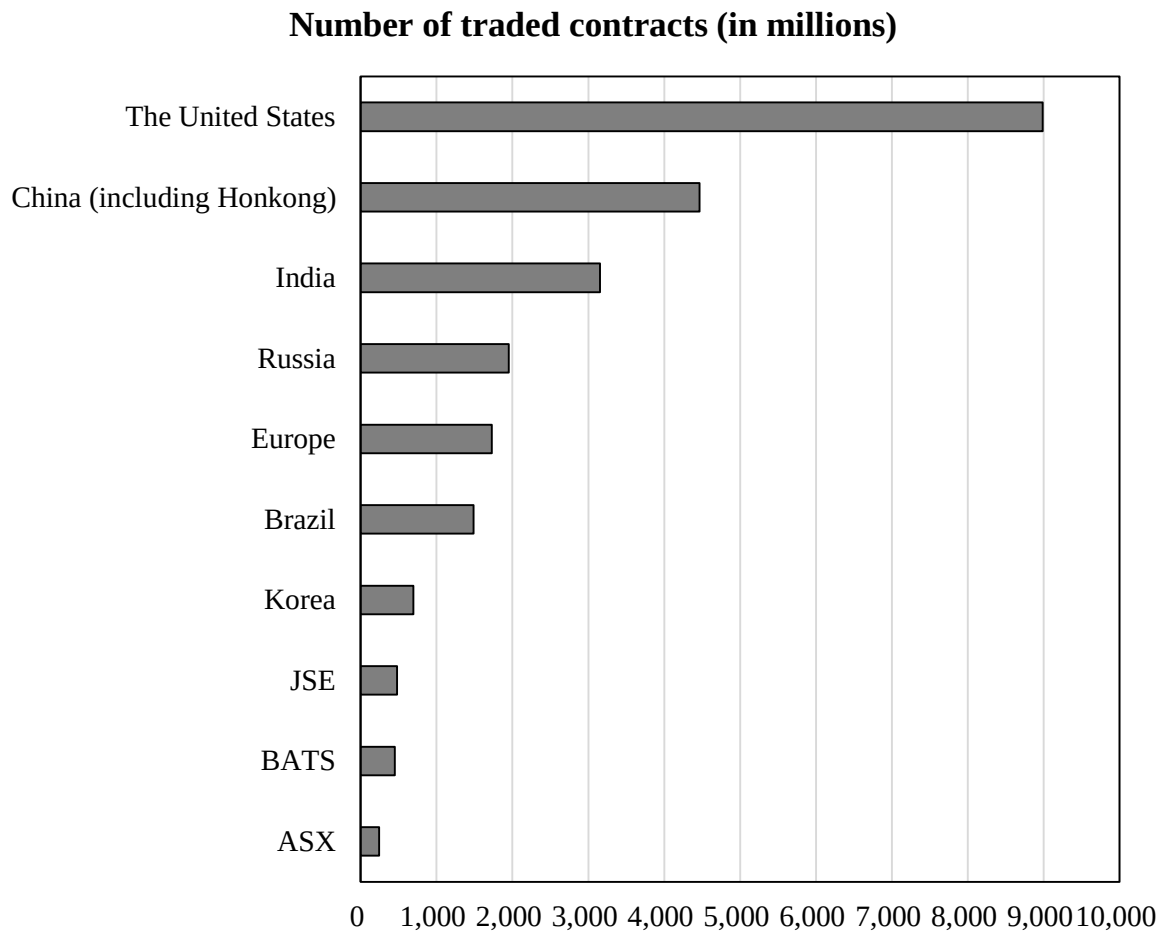
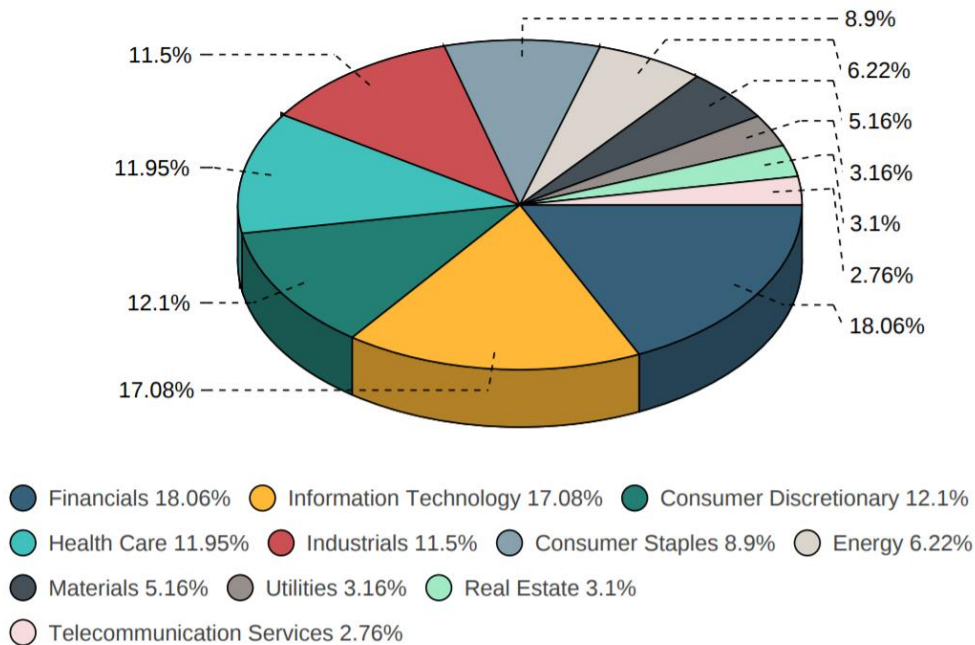


FIGURE 2 Sector and country weights of the MSCI index (MSCI, 2017).

The number of traded derivative contracts by country as of 2016 is reported. The data on the trading volumes of the top 20 exchanges around the world are collected from Statista and classified by country.

A. Sector weights



B. Country weights

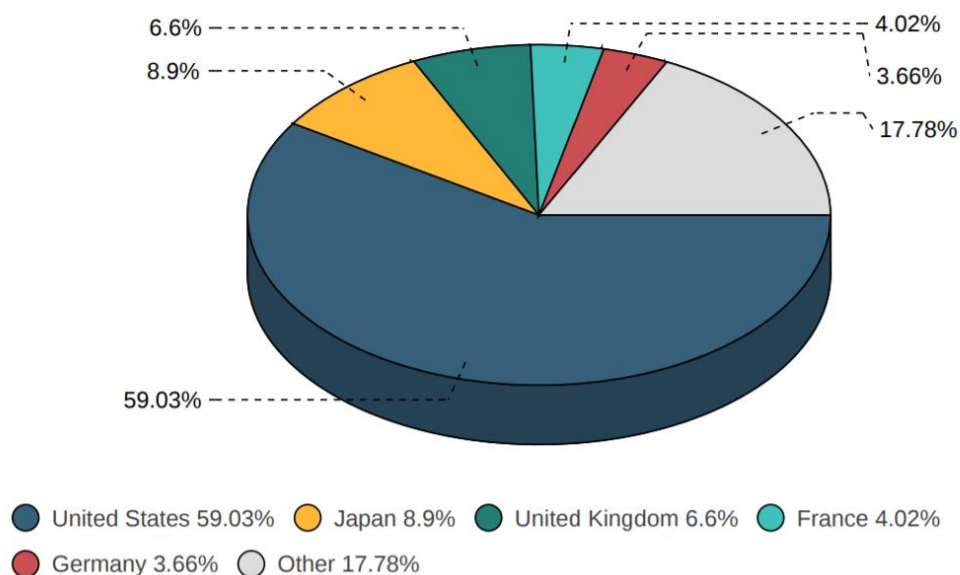


FIGURE 3 Monthly cumulative returns for Chinese commodity futures.

The daily closing prices of the generic first futures (in USD) between September 14, 2006, and October 31, 2017, are collected from Bloomberg. The cumulative returns from the first to the last day of the month are defined as the monthly returns.

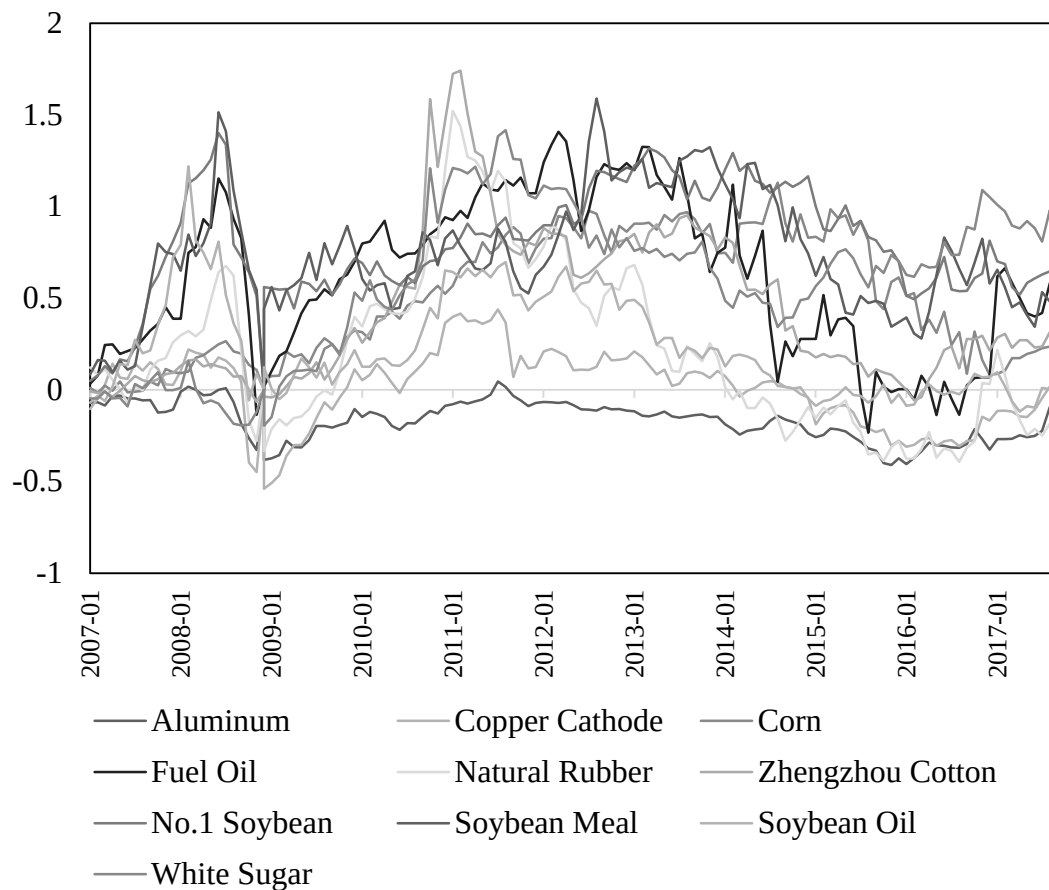
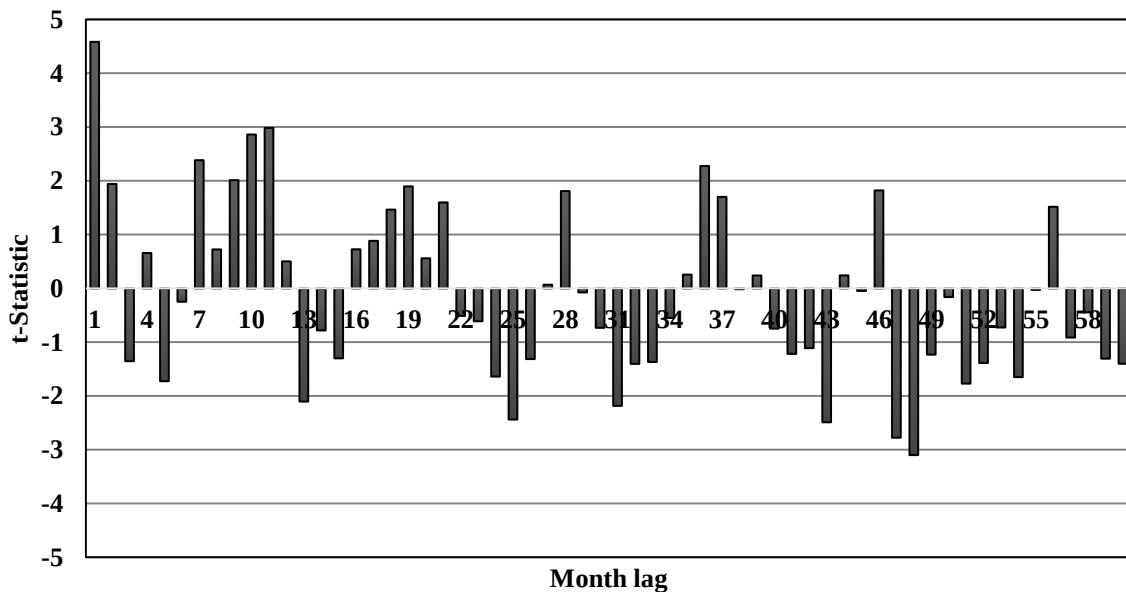


FIGURE 4 Time-series predictability for Chinese commodity futures.

The monthly excess returns of each futures contract, scaled by their ex ante volatilities, are regressed on their lagged scaled excess returns and the sign of the past excess return. Panel A indicates that the regression results use the excess returns themselves as dependent variables, and the dependent variables presented in Panel B indicate the signs of the lagged excess returns. Both regressions use lags from one and 60 months, and the sample period is September 2006 to October 2017. (A) Panel A: $r_t^s/\sigma_{t-1}^s = \alpha + \beta_h r_{t-h}^s/\sigma_{t-h-1}^s + \varepsilon_t^s$; (B) Panel B: $r_t^s/\sigma_{t-1}^s = \alpha + \beta_h \text{sign}(r_{t-h}^s) + \varepsilon_t^s$.

Panel A.



Panel B.

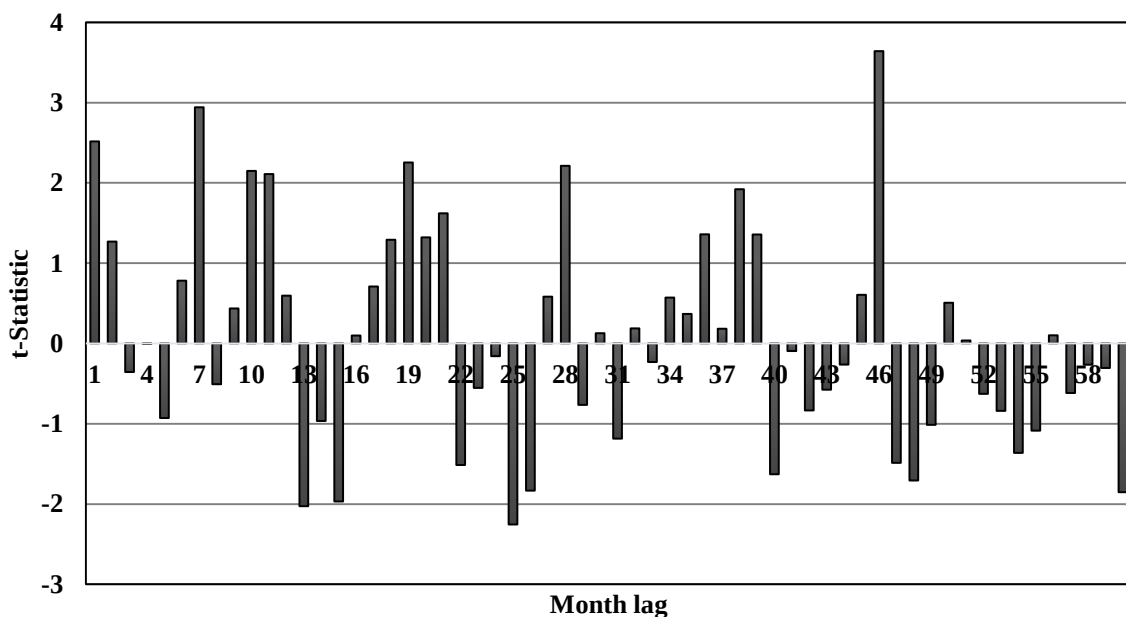
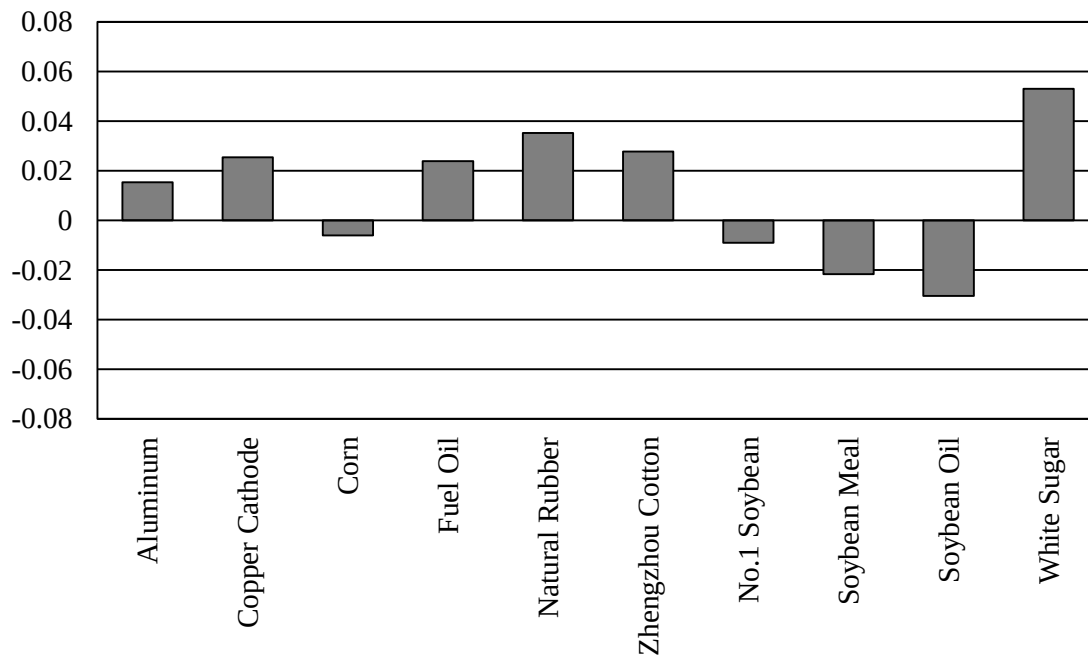
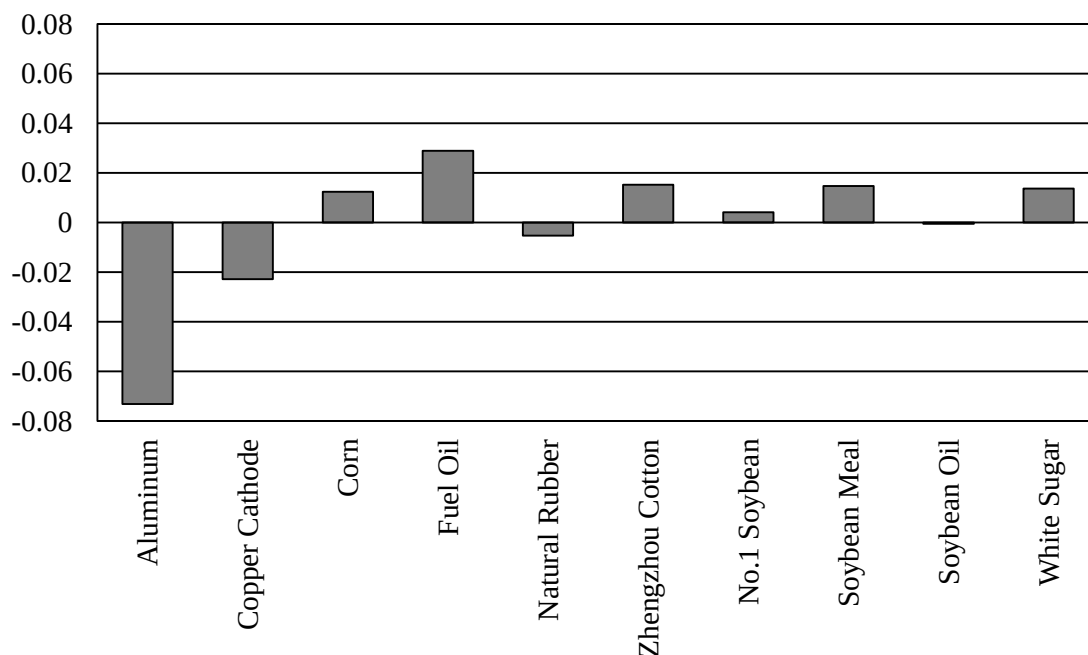


FIGURE 5 Annualized Sharpe ratios, by commodity, in the Chinese commodity futures market. This figure reports the annualized Sharpe ratios of the time-series momentum strategy (A), passive long strategy (B), and cross-sectional momentum strategy (C) by commodity futures contract. The time-series momentum and cross-sectional momentum use a 12-month look-back period and a one-month holding period (i.e., $k = 12$ and $h = 1$). The passive long strategy is the long-only strategy with risk-scaled positions constructed using the same method as those of the time-series and cross-sectional momentum strategies. The sample period is January 2008 to October 2017.

A. Sharpe ratios of the time-series momentum strategy



B. Sharpe ratios of the passive long strategy



C. Sharpe ratios of the cross-sectional momentum strategy

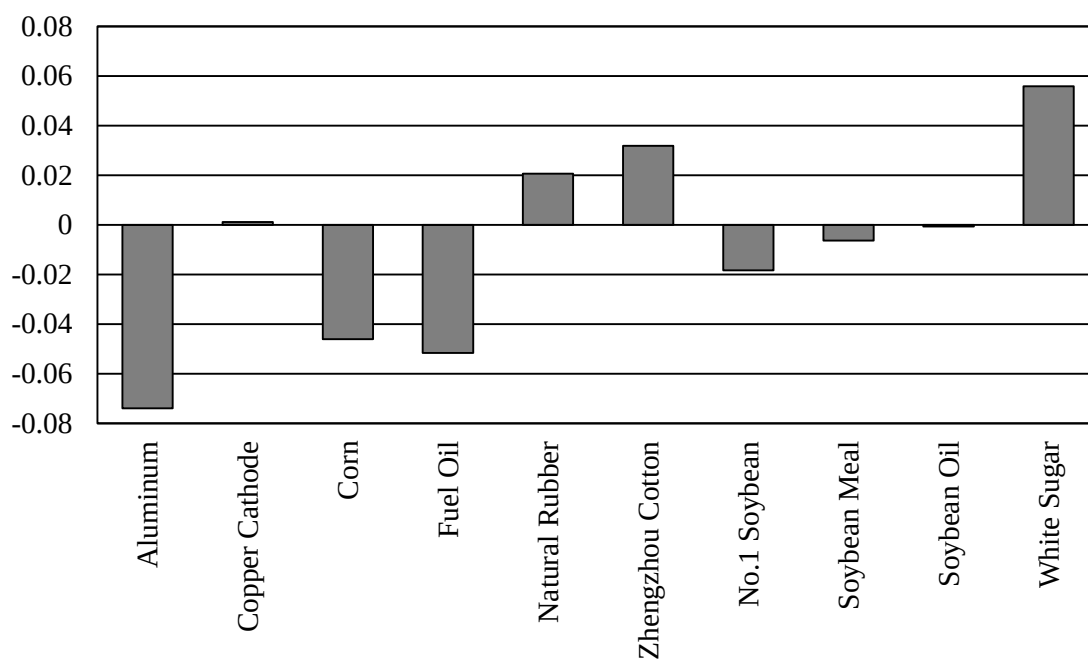


FIGURE 6 Cumulative returns of the time-series momentum strategy, passive long strategy, and cross-sectional momentum strategy.

TSMOM and XSMOM denote the time-series momentum and cross-sectional momentum, respectively, with a 12-month look-back period and one-month holding period (i.e., $k = 12$ and $h = 1$). The passive long strategy is the long-only strategy with risk-scaled positions, using the same method for TSMOM and XSMOM. The sample period is January 2008 to October 2017.

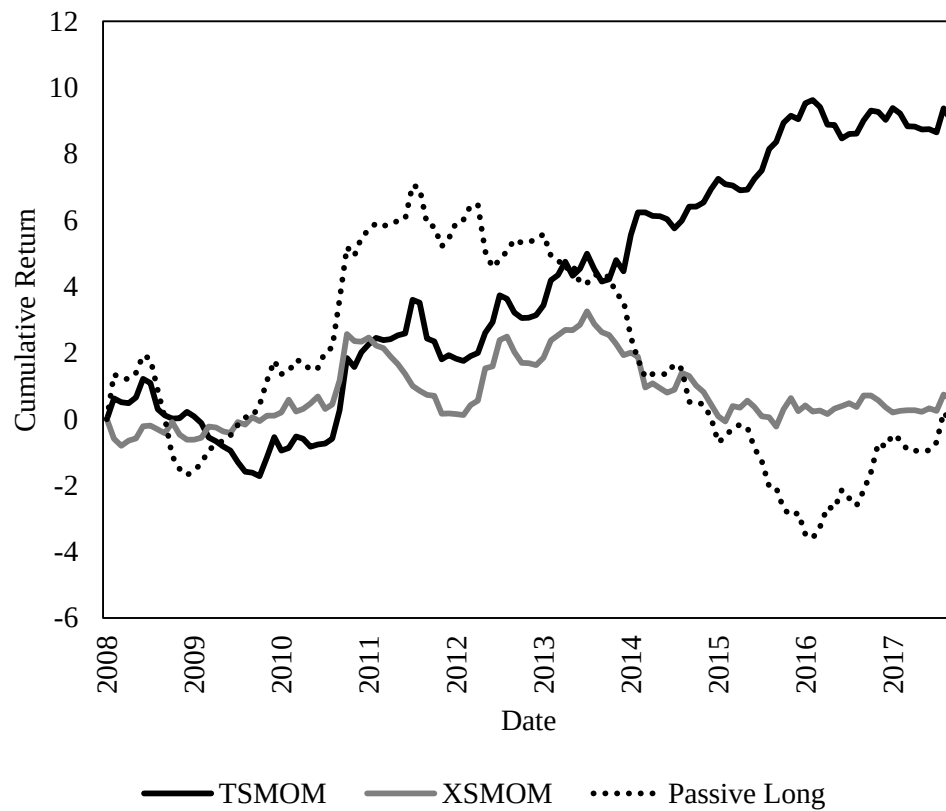


TABLE 1 List of commodity futures (Bloomberg ticker).

Aluminum Futures (AA), Copper Cathode Futures (CU), Corn Futures (AC), Fuel Oil Futures (FO), Natural Rubber Futures (RT), Zhengzhou Cotton Futures (VV), No.1 Soybean Futures (AK), Soybean Meal Futures (AE), Soybean Oil Futures (SH), and White Sugar Futures (CB)

Notes: This table lists the commodity futures analyzed in this study. The daily closing prices of the generic first futures (in USD) between September 14, 2006, and October 31, 2017, are collected from Bloomberg.

TABLE 2 Summary statistics of the daily excess returns on Chinese commodity futures.

Notes: This table reports the annualized arithmetic means and annualized volatilities from September 14, 2006, to October 31, 2017.

Commodity futures	Annualized mean	Annualized volatility
Aluminum	0.48%	14.60%
Copper cathode	1.58%	22.60%
Corn	6.42%	20.93%
Fuel oil	12.31%	37.15%
Natural rubber	0.58%	27.92%
Zhengzhou cotton	0.01%	16.89%
No.1 soybean	7.10%	18.68%
Soybean meal	9.16%	26.88%
Soybean oil	7.38%	30.41%
White sugar	8.69%	18.27%

TABLE 3 Correlations between the Chinese commodity futures.

Notes: This table reports the correlations between the 1-month returns of Chinese commodity futures from January 2007 to October 2017.

	Aluminum	Copper Cathode	Corn	Fuel Oil	Natural Rubber	Zhengzhou Cotton	No.1 Soybean	Soybean Meal	Soybean Oil	White Sugar
Aluminum	1.00									
Copper Cathode	0.56	1.00								
Corn	0.30	0.06	1.00							
Fuel Oil	0.21	0.31	-0.16	1.00						
Natural Rubber	0.35	0.56	0.06	0.34	1.00					
Zhengzhou Cotton	0.26	0.25	0.00	0.17	0.37	1.00				
No.1 Soybean	0.17	0.20	0.23	0.23	0.24	0.15	1.00			
Soybean Meal	0.23	0.31	0.17	0.17	0.28	0.17	0.43	1.00		
Soybean Oil	0.25	0.45	0.02	0.19	0.31	0.22	0.36	0.35	1.00	
White Sugar	0.25	0.32	-0.03	0.16	0.41	0.37	0.15	0.09	0.35	1.00

TABLE 4 The t-statistics of the alpha from the regressions of the time-series momentum strategy returns on the risk factors, with different look-back and holding periods.

Notes: The t-statistics of the intercepts are reported for the following regression: $r_t^{\text{TSMOM}(k,h)} = \alpha + \beta_1 \text{MKT}_t + \beta_2 \text{BOND}_t + \beta_3 \text{GSCI}_t + s\text{SMB}_t + h\text{HML}_t + m\text{UMD}_t + \varepsilon_t$, where MKT is the excess return on the MSCI World Index (Panel A) and CSI 300 Index (Panel B), BOND is the Barclay's Aggregate Bond Index, GSCI is the S&P GSCI Index, and the long-short factors (SMB, HML, and UMD) are the Fama–French factors for the size, value, and cross-sectional momentum premiums. The MSCI World Equity Index, CSI 300 Index, Barclay's Aggregate Bond Index, and S&P GSCI Index are obtained from Bloomberg, and the long-short factors are taken from Ken French's data library. The sample period is September 2006 to October 2017. The superscripts ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. MSCI World Index

		Holding period (months)							
		1	3	6	9	12	24	36	48
Lookback period (months)	1	2.60***	1.21	0.26	0.83	1.20	0.60	0.60	0.01
	3	1.41	0.50	0.17	0.64	0.77	0.52	-0.05	-0.65
	6	1.20	0.95	0.79	0.95	0.83	0.68	0.67	0.32
	9	1.77*	1.69*	1.44	1.27	1.29	1.23	1.07	0.61
	12	2.42**	1.66*	1.18	1.26	1.17	0.89	0.87	0.20
	24	1.42	1.21	1.01	0.94	0.85	0.64	0.06	-0.41
	36	0.68	0.80	0.72	0.39	0.48	0.03	-0.53	-0.63
	48	-1.28	-2.03	-2.17	-1.79	-1.71	-2.01	-2.33	-2.30

Panel B. CSI 300 Index

		Holding period (months)							
		1	3	6	9	12	24	36	48
Lookback period (months)	1	2.56**	1.30	0.11	0.59	0.69	0.37	0.43	-0.30
	3	1.47	0.41	0.14	0.41	0.47	0.45	-0.05	-0.74
	6	1.17	0.66	0.50	0.63	0.59	0.59	0.52	0.15
	9	1.70*	1.25	0.93	0.84	0.95	0.95	0.78	0.41
	12	1.98**	0.94	0.48	0.58	0.52	0.28	0.29	-0.38
	24	1.19	0.99	0.83	0.77	0.69	0.52	-0.02	-0.43
	36	0.45	0.60	0.52	0.20	0.29	-0.12	-0.65	-0.77

48	-1.36	-2.12	-2.24	-1.88	-1.80	-2.09	-2.44	-2.40
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TABLE 5 Regression of the returns from the time-series momentum strategy on the returns from the cross-sectional momentum strategy.

Notes: This table reports the results of the regression of the returns from TSMOM on the returns from XSMOM. TSMOM and XSMOM denote the time-series momentum and cross-sectional momentum strategies, respectively, with a 12-month look-back period and one-month holding period (i.e., $k = 12$ and $h = 1$). The sample period is January 2008 to October 2017. The superscripts ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

			Independent variables		
Dependent variable	TSMOM	Coefficient	XSMOM	Intercept	R^2
		(<i>t</i> -Stat)	0.54*** (5.24)	0.07** (2.29)	0.20

TABLE 6 Regression analysis of monthly returns for each strategy (time-series momentum strategy, passive long strategy, and cross-sectional strategy) on the traditional risk factors.

Notes: The risk factors employed are the GSCI Index excess returns, SMB, HML, and UMD, on which the returns from the time-series momentum strategy are regressed. The GSCI Index excess returns represent the global commodity market, the SMB and HML are the Fama and French (1993) size and value factors, and the UMD is the Carhart (1997) momentum factor. The GSCI Index is collected from Bloomberg, and the long-short factors (SMB, HML, and UMD) are taken from Ken French's data library. Panels A, B, and C represent the results of regressions for the time-series momentum strategy, passive long strategy, and cross-sectional momentum strategy, respectively. The sample period is January 2008 to October 2017. The superscripts ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

A-1. Regression analysis for the time-series momentum strategy

	GSCI	SMB	HML	UMD	Intercept	R ²
Coefficient	0.91*	0.01	-0.01	0.01***	0.20**	0.10
(t-Stat)	(1.69)	(0.20)	(-0.47)	(2.97)	(2.33)	

A-2. Regression analysis of monthly returns, other than the agricultural commodities, for the time-series momentum strategy on the traditional risk factors

	GSCI	SMB	HML	UMD	Intercept	R ²
Coefficient	0.80	-0.01	-0.02	0.03***	0.16***	0.08
(t-Stat)	(1.02)	(0.30)	(-0.60)	(2.58)	(3.11)	

B. Regression analysis for the passive long strategy

	GSCI	SMB	HML	UMD	Intercept	R ²
Coefficient	3.81***	0.02	-0.01	0.00	0.00	0.28
(t-Stat)	(6.25)	(0.72)	(-0.39)	(0.10)	(0.04)	

C. Regression analysis for the cross-sectional momentum strategy

	GSCI	SMB	HML	UMD	Intercept	R ²
Coefficient	0.48	-0.01	-0.02	0.00	0.01	0.03
(t-Stat)	(1.06)	(-0.44)	(-1.37)	(0.55)	(0.81)	

TABLE 7 Correlations between the returns of each Chinese futures contract in the time-series momentum strategy and passive long strategy.

Notes: The time-series momentum and cross-sectional momentum strategies use a 12-month look-back period and one-month holding period (i.e., $k = 12$ and $h = 1$). The passive long strategy is the long-only strategy with risk-scaled positions, constructed using the same method as the time-series and cross-sectional momentum strategies. The sample period for is January 2008 to October 2017.

A. Correlations of returns for the time-series momentum strategy

	Aluminum	Copper Cathode	Corn	Fuel Oil	Natural Rubber	Zhengzhou Cotton	No.1 Soybean	Soybean Meal	Soybean Oil	White Sugar
Aluminum	1.00									
Copper Cathode	0.15	1.00								
Corn	0.02	-0.19	1.00							
Fuel Oil	-0.05	-0.19	0.30	1.00						
Natural Rubber	-0.19	-0.20	-0.08	0.00	1.00					
Zhengzhou Cotton	-0.07	-0.07	-0.13	-0.08	0.34	1.00				
No.1 Soybean	0.00	-0.05	0.06	0.17	0.07	-0.07	1.00			
Soybean Meal	-0.10	-0.07	-0.09	0.02	0.15	-0.02	0.19	1.00		
Soybean Oil	-0.16	-0.16	0.09	0.18	0.14	0.06	0.12	0.12	1.00	
White Sugar	-0.08	-0.03	0.17	0.06	0.33	0.16	0.05	0.11	0.19	1.00

B. Correlations of returns for the passive long strategy.

	Aluminum	Copper Cathode	Corn	Fuel Oil	Natural Rubber	Zhengzhou Cotton	No.1 Soybean	Soybean Meal	Soybean Oil	White Sugar
Aluminum	1.00									
Copper Cathode	0.44	1.00								
Corn	0.10	-0.02	1.00							
Fuel Oil	0.04	0.28	0.02	1.00						
Natural Rubber	0.33	0.53	0.15	0.30	1.00					
Zhengzhou Cotton	0.20	0.21	0.09	0.15	0.38	1.00				
No.1 Soybean	0.12	0.25	0.18	0.24	0.18	0.23	1.00			
Soybean Meal	0.24	0.25	0.13	0.15	0.22	0.22	0.33	1.00		
Soybean Oil	0.17	0.28	0.02	0.32	0.36	0.19	0.39	0.33	1.00	
White Sugar	0.25	0.21	-0.03	0.22	0.44	0.23	0.08	0.01	0.31	1.00